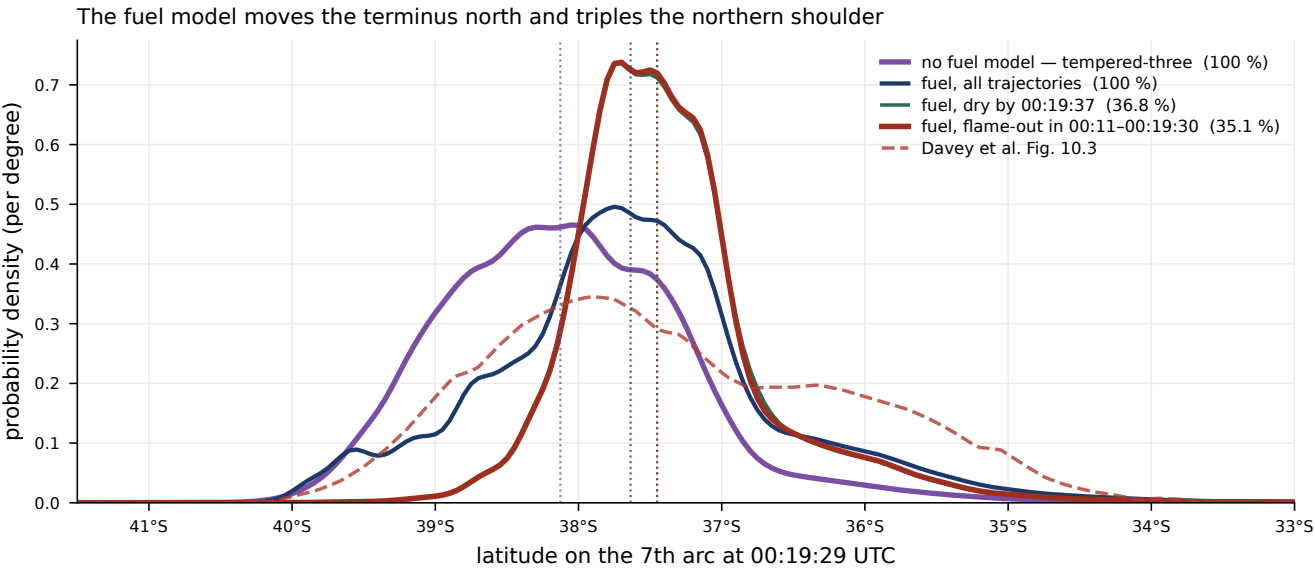


What the fuel model does: 6temper-realloc against the most convergent fuel-free run

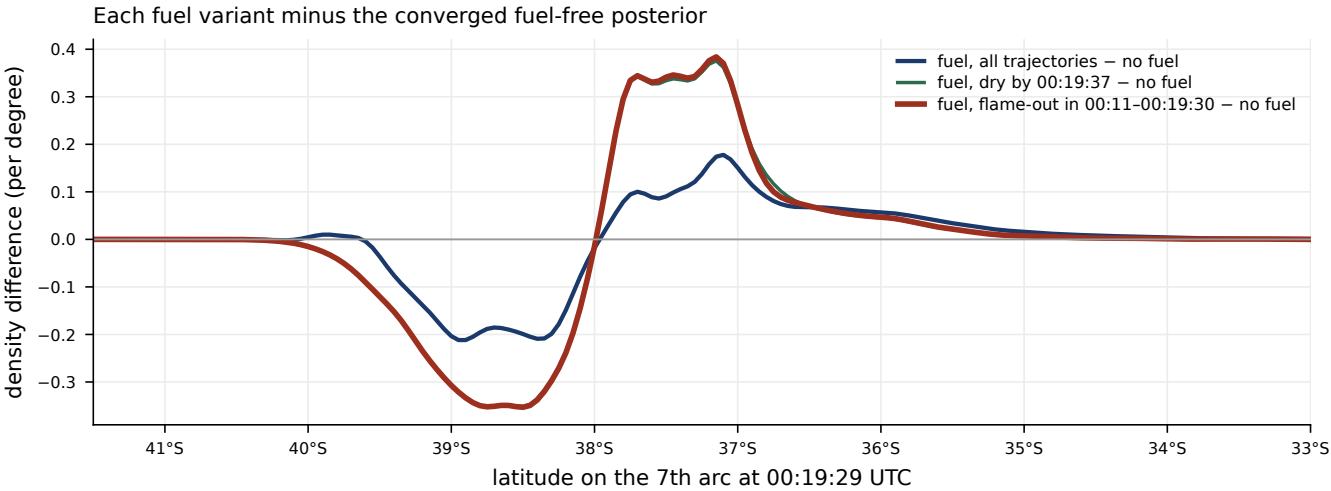
a



no fuel model — tempered-three	median -38.127°	50 % HDI width 1.15°	shoulder 0.0353 (14.7 % of Davey)	split-half 0.9551	PASS
fuel, all trajectories	median -37.636°	50 % HDI width 1.10°	shoulder 0.1042 (43.3 % of Davey)	split-half 0.9109	fail
fuel, dry by 00:19:37	median -37.452°	50 % HDI width 0.70°	shoulder 0.0843 (35.1 % of Davey)	split-half 0.9053	fail
fuel, flame-out in 00:11-00:19:30	median -37.451°	50 % HDI width 0.70°	shoulder 0.0859 (35.7 % of Davey)	split-half 0.9028	fail

All four curves are the mean of their own run's per-replicate densities — the same pooling the split-half statistic uses — so they are built identically and are directly differenceable. Medians are read off the 0.05° grid with the cumulative taken at each bin's upper edge. That estimator is exact: applied to the engine's own stratum-pooled density it returns the engine's raw-particle median to four decimals (tempered-three -38.1319, 6temper-realloc -37.6427) and Davey's Fig. 10.3 median as -37.5323. Replicate-mean pooling sits 0.005-0.006° north of stratum pooling, which is why the table reads -38.127 and -37.636 rather than -38.132 and -37.643.

b



fuel, all trajectories	total absolute difference from no-fuel	0.4754
fuel, dry by 00:19:37	total absolute difference from no-fuel	0.8578
fuel, flame-out in 00:11-00:19:30	total absolute difference from no-fuel	0.8610

no fuel to flame-out-conditioned: median moves +0.676° (41 NM). That is 16 times the median's own half-to-half agreement of 0.043°, but SMALLER than the 50 % HDI widths of 1.15° and 1.10°, so it is a shift in central location that the individual runs' 50 % intervals still straddle. The no-fuel run is the only series that clears the 0.924 convergence floor.